

PRACTICAL SESSION 01

Flutter for Beginners

Dart Fundamentals + Setting up Flutter

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A G E N D A

What we'll cover today

01

Dart Fundamentals

Variables, functions, control flow, collections, null safety, classes

02

Flutter Installation

Install Flutter SDK on macOS and Windows

03

IDE Setup

VS Code & Android Studio with Flutter plugins

04

First Project

Run flutter doctor and build a starter app



PART ONE

Dart Fundamentals

The language that powers Flutter

What is Dart?

Dart is the programming language used to build Flutter apps.

- **Strongly typed** Catches errors before your app runs
- **Object-oriented** Everything is an object, even numbers
- **Compiled** Fast on devices, hot-reload in dev
- **Made by Google** Built specifically with Flutter in mind

HELLO WORLD IN DART

```
void main() {  
  print('Hello, Flutter!');  
}
```

TRY IT ONLINE

dartpad.dev

Run Dart in your browser — no install needed

Variables & Data Types

COMMON TYPES

int	whole numbers	42
double	decimals	3.14
String	text	'Hello'
bool	true / false	true
var	type inferred	var x = 10;
final	set once	final pi = 3.14;
const	compile-time const	const max = 100;

EXAMPLE

```
// Explicit types
int age = 25;
String name = 'Muhammad';
double height = 1.75;
bool isStudent = true;

// Type inference with var
var city = 'Erbil';

// Cannot be reassigned
final country = 'Iraq';
const courseCode = 'FL101';
```

Functions

Reusable blocks of code that do one job.

STANDARD FUNCTION

```
int add(int a, int b) {  
  return a + b;  
}
```

```
add(2, 3); // 5
```

ARROW FUNCTION

```
int square(int x) => x * x;  
  
square(5); // 25  
  
// One-liner shorthand
```

NAMED PARAMETERS

```
String greet({  
  required String name,  
  String? title,  
}) => 'Hi $title $name';  
  
greet(name: 'Hama');
```

Making Decisions & Repeating

IF / ELSE

```
int score = 85;

if (score >= 90) {
  print('A');
} else if (score >= 80) {
  print('B');
} else {
  print('Keep going!');
}
```

FOR LOOP

```
for (int i = 1; i <= 5; i++) {
  print('Day $i');
}
```

WHILE LOOP

```
int count = 3;
while (count > 0) {
  print(count);
  count--;
}
```

Lists, Maps & Sets

LIST

Ordered, indexed

```
List<String> students = [  
  'Hama', 'Sara', 'Ali',  
];  
  
students.add('Lara');  
print(students[0]); // Hama  
print(students.length); // 4
```

MAP

Key → value pairs

```
Map<String, int> grades = {  
  'Hama': 92,  
  'Sara': 88,  
};  
  
grades['Ali'] = 75;  
print(grades['Hama']); // 92
```

SET

Unique items, no order

```
Set<String> tags = {  
  'flutter', 'dart',  
  'mobile',  
};  
  
tags.add('dart'); // ignored  
print(tags.contains('dart'));
```


Null Safety

Dart helps you avoid the dreaded null reference errors.

NON-NULLABLE BY DEFAULT

```
String name = 'Hama'; // OK
name = null;          // ERROR

String? nick;          // ? = nullable
nick = null;           // OK
nick = 'H';            // OK
```

SAFE OPERATORS

```
String? name;

// ?. – call only if not null
print(name?.length);

// ?? – default value
print(name ?? 'Guest');

// ! – assert not null (careful!)
print(name!.length);
```

Classes & Objects

In Flutter, everything is built from classes — widgets, screens, models.

```
class Student {  
  String name;  
  int grade;  
  
  Student(this.name, this.grade);  
  
  void greet() {  
    print('Hi, I am $name');  
  }  
}  
  
void main() {  
  var s = Student('Hama', 92);  
  s.greet();  
}
```

KEY IDEAS

Class

Blueprint for an object

Constructor

Creates a new instance

Property

Data the object holds

Method

Action the object can do



PART TWO

Installing Flutter

Get your machine ready to build apps

What You'll Need

Disk Space

At least 10 GB free for SDK + tools

Git

Required by the Flutter SDK

An IDE

VS Code or Android Studio

A Device

Real phone or emulator/simulator

Install Flutter on macOS

1

Install Xcode

From the Mac App Store. Required for iOS builds.

2

Install Homebrew

Package manager — makes the rest easy.

3

Install Flutter SDK

Run: `brew install --cask flutter`

4

Add to PATH

Edit `~/.zshrc` to expose flutter & dart commands

5

Accept Xcode license

Run: `sudo xcodebuild -license`

TERMINAL

```
# Install Homebrew
/bin/bash -c "$(curl -fsSL \
  https://raw.githubusercontent.com/\
  Homebrew/install/HEAD/install.sh)"

# Install Flutter
brew install --cask flutter

# Add to PATH (in ~/.zshrc)
export PATH="$PATH:`flutter`"

# Reload shell
source ~/.zshrc

flutter --version
```

Install Flutter on Windows

1

Download Flutter SDK

Get the .zip from flutter.dev/docs/get-started

2

Extract to C:\

Avoid spaces — don't use Program Files

3

Add to PATH

Add C:\flutter\bin to environment vars

4

Install Git for Windows

Required by the Flutter tooling

5

Install Android Studio

For Android SDK + emulator

POWERSHELL

```
# After extracting to C:\src

# Verify install
C:\> flutter --version

# If not recognized:
# 1. Win + R → sysdm.cpl
# 2. Advanced → Environment Vars
# 3. Edit Path → Add:
#    C:\src\flutter\bin
# 4. Restart terminal

flutter doctor
```

Pick Your Editor

Both work great. VS Code is lightweight; Android Studio is full-featured.

RECOMMENDED

VS Code

Fast, lightweight, huge extension ecosystem.

- Install from code.visualstudio.com
- Open Extensions (Ctrl/Cmd + Shift + X)
- Install: Flutter (by Dart Code)
- Dart extension installs automatically

FULL IDE

Android Studio

Built-in emulator manager and visual tools.

- Install from developer.android.com/studio
- Open Settings → Plugins
- Search & install: Flutter plugin
- Restart, then run AVD Manager

VERIFY

Run flutter doctor

This command checks every part of your setup and tells you what's missing.

TERMINAL OUTPUT

```
$ flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, on macOS)
[✓] Android toolchain - develop for Android devices
[✓] Xcode - develop for iOS and macOS
[✓] Chrome - develop for the web
[✓] VS Code
[✓] Connected device (1 available)

• No issues found!
```

Yellow !

Optional — only if you need that platform

Create & Run Your First App

COMMANDS

```
# 1. Create a new app
flutter create my_first_app

# 2. Move into the folder
cd my_first_app

# 3. List connected devices
flutter devices

# 4. Run it!
flutter run

# Hot reload while running:
# Press 'r' to reload
# Press 'R' to restart
# Press 'q' to quit
```

WHAT YOU GET

lib/main.dart

Your app's entry point

pubspec.yaml

Dependencies & assets

android/

Android-specific code

ios/

iOS-specific code

test/

Where your tests live

W R A P U P

What we covered today

Y O U N O W K N O W

- ✓ Dart syntax & data types
- ✓ Functions and control flow
- ✓ Lists, Maps, and Sets
- ✓ Null safety basics
- ✓ Classes & objects
- ✓ Flutter SDK installation
- ✓ IDE setup & flutter doctor

N E X T S E S S I O N

- Widgets — the heart of Flutter
- Stateless vs Stateful widgets
- Building your first UI
- Hot reload in action
- Layouts: Row, Column, Container
- Handling user input
- Mini project — a simple app



Thank U!

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